

E8H7C7

In the above situations 1 and 2, the total number of times the rectangular biscuit and the fidget spinner matches exactly with itself in one complete rotation is 2 and 3. This is called the ***order of rotational symmetry.*** In situation 3, the set square matches itself only once in one complete rotation and hence has no rotational symmetry.

An object is said to have a ***rotational symmetry*** if it looks the same after being rotated about its centre through an angle less than 360° (If the order of rotation of an object is atleast two).

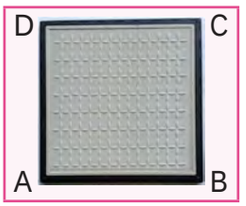
Think

Can you identify the object which does not have rotational symmetry in the above situations? Why?

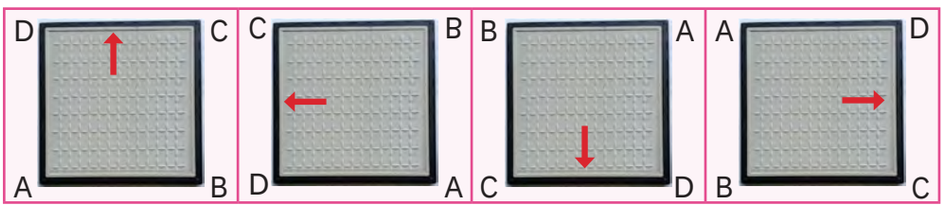
Example 10 A man-hole cover of a water sump is in square shape.

i) In how many ways we can fix that to close the sump?

ii) What is its order of rotational symmetry?



Solution



i) We can fix it in 4 ways as shown above.

ii) The order of rotational symmetry is 4.

Think

Suppose, the man hole cover of the water sump is in circular shape.



i) The number of ways to close that circular lid is _____

ii) What is its order of rotational symmetry?

Activity

Find the order of rotational symmetry by fixing the relevant shape in different ways.



6th_ Maths _Chapter 4.indd 64 14-10-2019 20:09:58